

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396604
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Conrad; Wayne Ernest

Configuration of a cyclone assembly and surface cleaning apparatus having same

Abstract

A cyclone separator useable in a surface cleaning apparatus comprises a cyclone chamber and a dirt collection chamber exterior to, and surrounding at least a portion of the cyclone chamber. The dirt collection chamber is in communication with the cyclone chamber via a dirt outlet. An air flow passage extending to the cyclone air inlet travels generally axially through the dirt collection chamber.

Inventors:	Conrad; Wayne Ernest (Hampton, CA)
Applicant:	Omachron Intellectual Property Inc. (Hampton, CA)
Family ID:	1000008781773
Assignee:	Omachron Intellectual Property Inc. (Hampton, CA)
Appl. No.:	18/076883
Filed:	December 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230165422 A1	Jun. 01, 2023

Related U.S. Application Data

continuation parent-doc US 16377778 20190408 US 11547259 child-doc US 18076883
continuation parent-doc US 14529027 20141030 US 10327612 20190625 child-doc US 16377778
continuation parent-doc US 13763477 20130208 US 8898857 20141202 child-doc US 14529027
division parent-doc US 12338022 20081218 ABANDONED child-doc US 13763477
us-provisional-application US 61014983 20071219

Publication Classification

Int. Cl.: **A47L9/16** (20060101); **B04C5/187** (20060101); B01D45/16 (20060101); B04C5/04 (20060101); B04C5/103 (20060101); B04C5/14 (20060101); B04C5/26 (20060101); B04C5/28 (20060101)

U.S. Cl.:

CPC **A47L9/1608** (20130101); **A47L9/1658** (20130101); **A47L9/1683** (20130101); **B04C5/187** (20130101); A47L9/1625 (20130101); A47L9/1633 (20130101); A47L9/1641 (20130101); A47L9/165 (20130101); B01D45/16 (20130101); B04C5/04 (20130101); B04C5/103 (20130101); B04C5/14 (20130101); B04C5/26 (20130101); B04C5/28 (20130101); Y10S55/03 (20130101)

Field of Classification Search

CPC: A47L (9/1608); A47L (9/1658); A47L (9/1683); A47L (9/165)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1779761	12/1929	Alford, Sr.	N/A	N/A
4373228	12/1982	Dyson	15/352	A47L 5/32
4790865	12/1987	Demarco	N/A	N/A
5139652	12/1991	Leblanc	N/A	N/A
5267371	12/1992	Soler et al.	N/A	N/A
5358290	12/1993	Fleet et al.	N/A	N/A
5391051	12/1994	Sabatier et al.	N/A	N/A
5681450	12/1996	Chitnis et al.	N/A	N/A
5922093	12/1998	James et al.	N/A	N/A
5974625	12/1998	Garner	N/A	N/A
6098244	12/1999	Conrad et al.	N/A	N/A
6099661	12/1999	Conrad	N/A	N/A
6101672	12/1999	Conrad et al.	N/A	N/A
6123779	12/1999	Conrad et al.	N/A	N/A
6129775	12/1999	Conrad et al.	N/A	N/A
6135708	12/1999	Conrad et al.	N/A	N/A
6168716	12/2000	Conrad et al.	N/A	N/A
6170119	12/2000	Conrad et al.	N/A	N/A
6171356	12/2000	Twerdun	N/A	N/A
6173474	12/2000	Conrad	N/A	N/A
6174127	12/2000	Conrad et al.	N/A	N/A
6183641	12/2000	Conrad et al.	N/A	N/A
6224325	12/2000	Conrad et al.	N/A	N/A
6231645	12/2000	Conrad et al.	N/A	N/A
6238177	12/2000	Conrad et al.	N/A	N/A
6243917	12/2000	Conrad	N/A	N/A
6251296	12/2000	Conrad et al.	N/A	N/A

6261052	12/2000	Conrad et al.	N/A	N/A
6261379	12/2000	Conrad et al.	N/A	N/A
6280143	12/2000	Parker et al.	N/A	N/A
6307358	12/2000	Conrad	N/A	N/A
6312594	12/2000	Conrad et al.	N/A	N/A
6328527	12/2000	Conrad et al.	N/A	N/A
6334234	12/2001	Conrad et al.	N/A	N/A
6344064	12/2001	Conrad	N/A	N/A
6398834	12/2001	Oh	N/A	N/A
6406505	12/2001	Oh et al.	N/A	N/A
6432154	12/2001	Oh et al.	N/A	N/A
6440197	12/2001	Conrad et al.	N/A	N/A
6457205	12/2001	Conrad	N/A	N/A
6532620	12/2002	Oh	N/A	N/A
6533834	12/2002	Conrad et al.	N/A	N/A
6599340	12/2002	Conrad et al.	N/A	N/A
6706095	12/2003	Morgan	N/A	N/A
6782585	12/2003	Conrad et al.	N/A	N/A
6810558	12/2003	Lee	55/459.3	A47L 9/1658
6868578	12/2004	Kasper et al.	N/A	N/A
7065826	12/2005	Arnold	N/A	N/A
7086119	12/2005	Go et al.	N/A	N/A
7160346	12/2006	Park	N/A	N/A
7329294	12/2007	Conrad	N/A	N/A
7419521	12/2007	Oh et al.	N/A	N/A
7422615	12/2007	Kim	55/459.1	A47L 9/1683
7511224	12/2008	Kossak	N/A	N/A
7544224	12/2008	Tanner et al.	N/A	N/A
7547351	12/2008	Oh	95/271	B01D 45/16
7618470	12/2008	Eddington et al.	N/A	N/A
7645309	12/2009	Jeong et al.	N/A	N/A
7736408	12/2009	Bck et al.	N/A	N/A
7740675	12/2009	Conrad	N/A	N/A
7744667	12/2009	Oh et al.	N/A	N/A
7749293	12/2009	Conrad	N/A	N/A
7776120	12/2009	Conrad	N/A	N/A
7803207	12/2009	Conrad	N/A	N/A
7811345	12/2009	Conrad	N/A	N/A
7867308	12/2010	Conrad	N/A	N/A
7887612	12/2010	Conrad	N/A	N/A
7941895	12/2010	Conrad	N/A	N/A
8034140	12/2010	Conrad	N/A	N/A
8127398	12/2011	Conrad	N/A	N/A
8146201	12/2011	Conrad	N/A	N/A
8151407	12/2011	Conrad	N/A	N/A
8176596	12/2011	Conrad	N/A	N/A
8292979	12/2011	Conrad	N/A	N/A
8349428	12/2012	Conrad	N/A	N/A
8590102	12/2012	Conrad	N/A	N/A
8621709	12/2013	Conrad	N/A	N/A

8677558	12/2013	Conrad	N/A	N/A
8713751	12/2013	Conrad	N/A	N/A
8869344	12/2013	Conrad	N/A	N/A
9119513	12/2014	Conrad	N/A	N/A
10537219	12/2019	Conrad et al.	N/A	N/A
11793374	12/2022	Conrad	N/A	N/A
11857142	12/2023	Conrad	N/A	N/A
2001/0027587	12/2000	Conrad et al.	N/A	N/A
2002/0020154	12/2001	Yang	N/A	N/A
2002/0124538	12/2001	Oh et al.	N/A	N/A
2002/0134059	12/2001	Oh	N/A	N/A
2002/0162188	12/2001	Harmen	N/A	N/A
2003/0201754	12/2002	Conrad	N/A	N/A
2004/0060146	12/2003	Coates et al.	N/A	N/A
2004/0216266	12/2003	Conrad	N/A	N/A
2005/0138763	12/2004	Tanner et al.	N/A	N/A
2006/0137304	12/2005	Jeong et al.	N/A	N/A
2006/0196004	12/2005	Conrad	N/A	N/A
2007/0084160	12/2006	Kim	55/345	A47L 9/1625
2007/0084161	12/2006	Yoo	N/A	N/A
2007/0209339	12/2006	Conrad	N/A	N/A
2007/0262512	12/2006	Watanabe et al.	N/A	N/A
2007/0289267	12/2006	Makarov et al.	N/A	N/A
2008/0172992	12/2007	Conrad	N/A	N/A
2008/0172995	12/2007	Conrad	N/A	N/A
2008/0178416	12/2007	Conrad	N/A	N/A
2008/0178418	12/2007	Conrad	N/A	N/A
2008/0179133	12/2007	Conrad	N/A	N/A
2008/0184681	12/2007	Oh et al.	N/A	N/A
2008/0190080	12/2007	Oh et al.	N/A	N/A
2008/0196366	12/2007	Conrad	N/A	N/A
2008/0216281	12/2007	Conrad	N/A	N/A
2009/0181841	12/2008	Conrad	N/A	N/A
2009/0205161	12/2008	Conrad	N/A	N/A
2009/0209403	12/2008	Conrad	N/A	N/A
2011/0219570	12/2010	Conrad	N/A	N/A
2013/0145575	12/2012	Conrad	N/A	N/A
2014/0237768	12/2013	Conrad	N/A	N/A
2016/0367094	12/2015	Conrad	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1077412	12/1979	CA	N/A
2275675	12/1998	CA	N/A
2288544	12/1999	CA	N/A
2258416	12/1999	CA	N/A
2258419	12/1999	CA	N/A
2258421	12/1999	CA	N/A
2258422	12/1999	CA	N/A
2258423	12/1999	CA	N/A

2258426	12/1999	CA	N/A
2330801	12/2000	CA	N/A
2338193	12/2001	CA	N/A
2339514	12/2001	CA	N/A
2342673	12/2001	CA	N/A
2342993	12/2001	CA	N/A
2438079	12/2001	CA	N/A
1493244	12/2003	CN	N/A
1875846	12/2005	CN	N/A
1875855	12/2005	CN	N/A
1887437	12/2006	CN	N/A
100998484	12/2006	CN	N/A
101015436	12/2006	CN	N/A
101108111	12/2006	CN	N/A
101061932	12/2006	CN	N/A
101073480	12/2006	CN	N/A
101108106	12/2007	CN	N/A
101108110	12/2007	CN	N/A
101657133	12/2009	CN	N/A
69110424	12/1995	DE	N/A
10132690	12/2001	DE	N/A
60105004	12/2004	DE	N/A
102005015004	12/2007	DE	N/A
102005014541	12/2007	DE	N/A
602004009782	12/2007	DE	N/A
112007003039	12/2008	DE	N/A
112007003052	12/2009	DE	N/A
102008055045	12/2009	DE	N/A
202012101457	12/2011	DE	N/A
112012000251	12/2012	DE	N/A
102012223983	12/2013	DE	N/A
493950	12/1991	EP	N/A
1676516	12/2005	EP	N/A
1779761	12/2006	EP	N/A
2003135335	12/2002	JP	N/A
2010227287	12/2009	JP	N/A
1020050091821	12/2004	KR	N/A
1020050091824	12/2004	KR	N/A
1020050091826	12/2004	KR	N/A
1020050091829	12/2004	KR	N/A
1020050091830	12/2004	KR	N/A
1020050091833	12/2004	KR	N/A
1020050091834	12/2004	KR	N/A
1020050091835	12/2004	KR	N/A
1020050091836	12/2004	KR	N/A
1020050091837	12/2004	KR	N/A
1020050091838	12/2004	KR	N/A
1020050103343	12/2004	KR	N/A
1020050104613	12/2004	KR	N/A
1020050104614	12/2004	KR	N/A

1020060018004	12/2005	KR	N/A
1020060081229	12/2005	KR	N/A
1020060112420	12/2005	KR	N/A
1020060118795	12/2005	KR	N/A
1020060118800	12/2005	KR	N/A
1020060118801	12/2005	KR	N/A
1020060118802	12/2005	KR	N/A
1020060118803	12/2005	KR	N/A
1020060122249	12/2005	KR	N/A
1020080039105	12/2007	KR	N/A
9640840	12/1995	WO	N/A
0010718	12/1999	WO	N/A
2007021043	12/2006	WO	N/A
2007104138	12/2006	WO	N/A
2009076773	12/2008	WO	N/A
2009076774	12/2008	WO	N/A

OTHER PUBLICATIONS

English machine translation of KR1020080039105, published on May 7, 2008. cited by applicant
English machine translation of KR1020060122249, published on Nov. 30, 2006. cited by applicant
English machine translation of KR1020060118803, published on Nov. 24, 2008. cited by applicant
English machine translation of KR1020060118802, published on Nov. 24, 2006. cited by applicant
English machine translation of KR1020060118801, published on Nov. 24, 2006. cited by applicant
English machine translation of KR1020060118800, published on Nov. 24, 2006. cited by applicant
Nicholas P. Cheremisinoff: Handbook of Air Pollution Prevention and Control, N&P Limited, published by Butterworth Heinemann, pp. 397-404, Copyright 2002. Elsevier Science, ISBN 0-7506-7499-7. cited by applicant

International Preliminary Report on Patentability, received in connection to international Patent Application No. PCT/CA2008/002257, dated Jun. 22, 2010. cited by applicant

International Search Report, received in connection to International Patent Application No. PCT/CA2008/002257, dated Apr. 14, 2009. cited by applicant

International Preliminary Report on Patentability, received in connection to International Patent Application No. PCT/CA2008/002258, dated Jun. 22, 2010. cited by applicant

International Search Report, received in connection to International Patent Application No. PCT/CA2008/002258, dated Apr. 9, 2009. cited by applicant

Extended European Search Report received in connection to European Application No./Patent No. 08862927.4, dated Sep. 13, 2011. cited by applicant

English machine translation of KR1020060118795, published on Nov. 24, 2006. cited by applicant

English machine translation of KR1020060112420, published on Nov. 1, 2006. cited by applicant

English machine translation of KR1020060081229, published on Jul. 12, 2006. cited by applicant

English machine translation of KR1020060018004, published on Feb. 28, 2006. cited by applicant

English machine translation of KR1020050104614, published on Nov. 3, 2005. cited by applicant

English machine translation of KR1020050104613, published on Nov. 3, 2005. cited by applicant

English machine translation of KR1020050103343, published on Oct. 31, 2005. cited by applicant

English machine translation of KR1020050091838, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091837, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091836, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091835, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091834, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091833, published on Sep. 15, 2005. cited by applicant

English machine translation of KR1020050091830, published on Sep. 15, 2005. cited by applicant
English machine translation of KR1020050091829, published on Sep. 15, 2005. cited by applicant
English machine translation of KR1020050091826, published on Sep. 15, 2005. cited by applicant
English machine translation of KR1020050091824, published on Sep. 15, 2005. cited by applicant
English machine translation of KR1020050091821, published on Sep. 15, 2005. cited by applicant
English machine translation of JP2010227287, published on Oct. 14, 2020. cited by applicant
English machine translation of JP2003135335, published on May 13, 2003. cited by applicant
Engliksh machine translation of CN101657133, published on Feb. 24, 2010. cited by applicant
English machine translation of DE69110424, published on Jul. 20, 1995. cited by applicant
English machine translation of DE602004009782, published on Aug. 28, 2008. cited by applicant
English machine translation of DE202012101457, published on Aug. 16, 2012. cited by applicant
English machine translation of DE112012000251, published on Oct. 17, 2013. cited by applicant
English machine translation of DE112007003052, published on 2010-01-14. cited by applicant
English machine translation of DE112007003039, published on Oct. 29, Oct. 29, 2009. cited by applicant
English machine translation of DE102012223983, published on Jun. 26, 2014. cited by applicant
English machine translation of DE102008055045, published on Jun. 24, 2010. cited by applicant
English machine translation of DE102005015004, published on Feb. 7, 2008. cited by applicant
English machine translation of DE102005014541, published on Aug. 28, 2008. cited by applicant
English machine translation of CN101108111A, published on Aug. 20, 2007. cited by applicant
English machine translation of CN101108110A, published on Feb. 23, 2008. cited by applicant
English machine translation of CN101108106A, published on Jan. 23, 2008. cited by applicant
English machine translation of CN101073480A, published on Nov. 21, 2007. cited by applicant
English machine translation of CN101061932A, published on Oct. 31, 2007. cited by applicant
English machine translation of CN101015436A, published on Aug. 15, 2007. cited by applicant
English machine translation of CN100998484A, published on Jul. 18, 2007. cited by applicant
English machine translation of DE60105004, published on Aug. 18, 2005. cited by applicant
English machine translation of DE10132690, published on Jul. 18, 2002. cited by applicant
English machine translation of CN1887437, published on Jan. 3, 2007. cited by applicant
English machine translation of CN1875855, published on Dec. 13, 2006. cited by applicant
English machine translation of CN1875846, published on Dec. 13, 2006. cited by applicant
English machine translation of CN1493244, published on May 5, 2004. cited by applicant
U.S. Appl. No. 09/343,479, filed Jun. 30, 1999. cited by applicant

Primary Examiner: Muller; Bryan R

Attorney, Agent or Firm: SMART & BIGGAR LP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/377,778, which was filed on Apr. 8, 2019, now allowed, which itself is a continuation of U.S. patent application Ser. No. 14/529,027, which was filed on Oct. 30, 2014 and issued as U.S. Pat. No. 10,327,612 on Jun. 25, 2019, which itself is a continuation of U.S. patent application Ser. No. 13/763,477, which was filed on Feb. 8, 2013 and issued as U.S. Pat. No. 8,898,857 on Dec. 2, 2014, which itself is a divisional application of U.S. patent application Ser. No. 12/338,022, which was filed on Dec. 18, 2008, now abandoned, which itself claims the benefit

FIELD

(1) This invention relates to a cyclone separator. In a preferred embodiment, the invention relates to a cyclonic separator, or a plurality of cyclonic separators in parallel, utilized as a cleaning stage in a surface cleaning apparatus such as a vacuum cleaner.

BACKGROUND

(2) Cyclonic separators, including those used in vacuum cleaners are known in the art. Typically, a cyclonic separator has an inlet for fluid (air, liquid or and air and liquid mix) to be treated and an outlet for treated fluid. Dirt may be collected either in the cyclone chamber itself (e.g. in the bottom) or in a collection chamber in fluid communication with the cyclone separator. Various such constructions are known in the art.

(3) U.S. Pat. No. 7,086,119 (Go et al) discloses a dust-collecting unit for a vacuum cleaner. The dust-collecting unit includes a cyclone separator having a dirt collection chamber positioned adjacent one lateral side of the cyclone separator. A dirt outlet is provided in the upper wall of the cyclone such that dirt may enter the adjacent dirt collection chamber through the outlet in the upper wall of the cyclone separator. A second dirt collection chamber is positioned below the cyclone chamber and is accessed by an opening formed in a separating plate that separates the cyclone chamber and the second dirt collection chamber. An openable bottom is provided. However, when the bottom is opened, the cyclone chamber is still closed by the separating plate.

(4) U.S. Pat. No. 7,160,346 (Park) discloses a cyclone for use in a vacuum cleaner having a dirt collection space positioned below the cyclone chamber. A dirt outlet is provided as an annular gap between the sidewall of the cyclone chamber and a separating plate for permitting dirt to travel downwardly from the cyclone into the dirt collection chamber. Accordingly, the dirt collection chamber is not exterior to the cyclone casing but is within the casing.

SUMMARY

(5) In accordance with this invention, a cyclone separator is provided that comprises a fluid inlet provided in a first portion, a separated material outlet provided in a spaced apart, opposed second portion and being in communication with a separated material collection chamber and a side wall. The separated material collection chamber is spaced a sufficient distance from the cyclone casing (at least portions of the side and the opposed second portion) to enhance the separation efficiency of the cyclone separator. In particular, the separated material collection chamber is spaced from at least some of the cyclone casing by at least 0.5 inches.

(6) The separated material collection chamber may surround all or a portion of the cyclone casing and is exterior to the portion of the cyclone casing having the separated material outlet. The sidewall of the cyclone casing has an inlet section and a second section and the fluid inlet is provided on the inlet section. Accordingly, the sidewall of the separated material collection chamber facing the sidewall of the second section of the cyclone casing is spaced apart by at least about 0.5 inches and, the separated material collection chamber has an opposed surface facing the second portion of the cyclone casing by at least about 0.5 inches.

(7) Surprising, it has been determined that such positioning of the separated material collection chamber enhances the separation efficiency of the cyclone separator.

(8) The separated material collection chamber (e.g., a dirt collection chamber) may extend completely around the cyclone casing. Accordingly, fluid will enter a cyclone chamber through the fluid inlet at one end and travel towards the distal end wall of the cyclone chamber. As the fluid travels through the cyclone chamber, it will rotate and heavier material (e.g. particulate matter) will exit the cyclone chamber via the separated material outlet. The fluid will then reverse direction and exit the cyclone chamber through the fluid outlet.

(9) Preferably, the cyclone separator is provided with an inwardly directed transition member that

extends between the sidewall of the cyclone casing and an end wall of the cyclone casing. The transition member may extend at an angle or may be curved inwardly. If the cyclone separator is oriented in an upright fashion, then the fluid inlet is provided adjacent the upper end of the cyclone and the fluid exit may also be provided adjacent the upper end of the cyclone. Accordingly, the transition member would extend downwardly and inwardly from the sidewall to the lower end wall that is positioned distal to the fluid inlet. It will be appreciated that if the cyclone separator were to be inverted, then the fluid inlet would be positioned on the bottom and the end wall would be positioned above the inlet (e.g., the end wall would be an upper end wall). In such an orientation, the transition member will extend upwardly and inwardly from the sidewall to the end wall. It will also be appreciated that the cyclone separator could also be oriented horizontally, or at any other angle.

(10) Preferably, the separated material outlet is provided adjacent the end wall distal to the fluid inlet and, more preferably, the dirt outlet is provided at least partially in the transition member. Most preferably, the dirt outlet is provided in the transition member.

(11) In accordance with an aspect of the present invention, there is provided a cyclone separator comprising: (a) a cyclone casing defining a cyclone chamber and having first and second spaced apart portions, a sidewall, a fluid inlet, a fluid outlet and a separated material outlet, the first portion including a first end and the second portion including an opposed second end; (b) the fluid inlet provided in the first portion; (c) the separated material outlet provided in the second portion and being in communication with a separated material collection chamber; (d) the sidewall having an inlet section and a second section, the fluid inlet provided on the inlet section; (e) the separated material collection chamber having a sidewall and surrounding at least a portion of the cyclone chamber wherein the sidewall of the separated material collection chamber facing the sidewall of the second section of the cyclone casing is spaced apart by at least about 0.5 inches; and, (f) the separated material collection chamber has an opposed surface facing the second end and spaced from the second end by at least about 0.5 inches.

(12) It will be appreciated by those skilled in the art that the cyclone separator disclosed herein may be utilized with any fluid stream (e.g. liquid and/or gas). In addition, it will be appreciated by those skilled in the art that the cyclone separator may be used in any consumer appliance and, preferably, is utilized in a surface cleaning apparatus or an air cleaner. The surface cleaning apparatus may be a vacuum cleaner, including an upright vacuum cleaner, a stick vacuum cleaner, a canister vacuum cleaner, a back pack vacuum cleaner, a strap carryable vacuum cleaner or a portable vacuum cleaner; a carpet extractor, a bare floor cleaner or the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective vertical section through a cyclone separator according to a first embodiment of this invention wherein the cyclone separator is oriented in an upright fashion;
- (2) FIG. 1a is a vertical section through the cyclone separator of FIG. 1;
- (3) FIG. 1b is a perspective vertical section looking downwardly through the cyclone separator of FIG. 1;
- (4) FIG. 2 is a vertical perspective view as shown in FIG. 1 including the air flow and dirt flow path through the cyclone separator;
- (5) FIG. 3 is a side view of the vertical section of FIG. 1;
- (6) FIG. 4 is an exploded view of the cyclone separator of FIG. 1;
- (7) FIG. 5 is a vertical section through the exploded view of FIG. 4;
- (8) FIG. 6 is a perspective view showing an optional embodiment that permits the cyclone separator to be opened to permit emptying;

- (9) FIG. 7 is an end view of the opened cyclone separator shown in FIG. 6;
- (10) FIG. 8 is a vertical section through a cyclone separator according to a second embodiment of this invention wherein the cyclone separator is inverted;
- (11) FIG. 9 is a side view of the cyclone separator of FIG. 8 shown encased in a housing;
- (12) FIG. 10 is an exploded view of the second embodiment of FIG. 8;
- (13) FIG. 11 is a perspective view from below of the cyclone separator of FIG. 8 showing an optional embodiment that permits the cyclone separator to be opened to permit emptying;
- (14) FIG. 12 is an end view of the opened cyclone separator shown in FIG. 11;
- (15) FIG. 13 is a vertical section through a third embodiment of a cyclonic cleaning stage according to this invention wherein a plurality of cyclone separators are connected in parallel;
- (16) FIG. 14 is a front view of the vertical section of the cyclonic cleaning stage of FIG. 13;
- (17) FIG. 15 is a front view of a vertical section of a cyclonic cleaning stage according to a fourth embodiment of this invention wherein the transition member is angled;
- (18) FIG. 16 is a front view of a vertical section of a cyclonic cleaning stage according to a fifth embodiment of this invention wherein the transition member is angled and the cyclonic cleaning stage comprises a single cyclone;
- (19) FIG. 17 is a front view of a vertical section of a cyclonic cleaning stage according to a sixth embodiment of this invention wherein the transition member is rounded, the cyclonic cleaning stage comprises a single cyclone and the separated material outlet extends above the transition member and is positioned about 270° around the cyclone casing in a flow direction from the fluid inlet;
- (20) FIG. 18 is a front view of a vertical section of a cyclonic cleaning stage according to a seventh embodiment of this invention wherein the transition member is rounded, the cyclonic cleaning stage comprises a single cyclone and the separated material outlet extends above the transition member and is positioned opposed to the air inlet;
- (21) FIG. 19 is a front view of a vertical section of a cyclonic cleaning stage according to an eighth embodiment of this invention wherein the cyclone chamber is openable with the dirt collection chamber, showing the dirt collection chamber and the cyclone chamber in the closed position; and,
- (22) FIG. 20 is a front view of a vertical section of a cyclonic cleaning stage according to a eighth embodiment of this invention wherein the cyclone chamber is openable with the dirt collection chamber, showing the dirt collection chamber and the cyclone chamber in the open position;
- (23) FIG. 21 is a top plan view of a cyclone chamber and a dirt collection chamber according to a ninth embodiment of this invention; and,
- (24) FIG. 22 is a perspective view of a surface cleaning apparatus utilizing a cyclone separator assembly as setout herein

DETAILED DESCRIPTION

(25) In the following description of the preferred embodiment, the cyclone separator is described as used in a vacuum cleaner of any particular design. As exemplified in FIG. 22, surface cleaning apparatus 120 may be an upright vacuum cleaner having a surface cleaning head 122 and a vacuum cleaner body 124 pivotally mounted thereto. Handle 126 may be provided for moving surface cleaning apparatus 120. Surface cleaning apparatus 120 has a first cyclonic cleaning stage 128 and a second cyclonic cleaning stage 130. However, it will be appreciated that the description set out herein is not confined to such uses as may be used for any other application referred to herein or known in the art.

(26) As exemplified in FIGS. 1-5, a cyclone separator assembly 10 comprises a cyclone casing defining a cyclone chamber 26 and comprising a sidewall 12, air or fluid inlet 18, air or fluid outlet 20, optional transition member 22 and a dirt or separated material outlet 24. The cyclone casing has first and second spaced apart portions. The first portion comprises first end wall 14 and the second portion comprises an opposed second end wall 16. A separated material collection chamber or dirt collection chamber 32 is provided exterior to cyclone chamber 26.

(27) First and second end walls **14** and **16** are spaced apart and are positioned opposite each other. Preferably, as exemplified, air inlet **18** is provided in sidewall **14** and, more preferably, adjacent first or upper end **14**. In addition, air outlet **20** is preferably provided in upper first end **14** and, preferably, is centrally located therein. As exemplified in FIG. **14**, the lower end of outlet **20** is preferably positioned above the top of dirt outlet **24**. Accordingly, as exemplified in a vertical orientation in FIG. **2**, air entering the cyclone casing will travel in a cyclonic fashion downwardly towards second lower end wall **16**. Heavier material, e.g. particulate material, will exit cyclone chamber **26** via dirt outlet **24**. The air at some point reverses direction and travels upwardly through outlet **20** to exit cyclone chamber **26**.

(28) As shown in FIG. **1**, sidewalls **18** preferably extend linearly (i.e. it is straight) and, in the orientation shown in FIG. **1**, vertically. Preferably, sidewall **18** meets upper end wall **14** at about 90° such that upper first end wall **14** is essentially perpendicular to sidewall **18**. In addition, lower second end wall **16** is preferably parallel to first end wall **14**. Accordingly, but for transition member **22**, the cyclone casing is exemplified as being cylindrical. It will be appreciated that while cyclone separator **10** is preferably generally cylindrical, it may have other shapes. For example, it may be frustoconical as is also known in the art. In addition, air inlet **18** and air outlet **20** may be of any construction and positioning known in the art.

(29) Transition member **12** is provided adjacent lower end wall **16** and preferable is immediately adjacent lower end wall **16**. Accordingly, transition member **22** may link sidewall **18** and lower end wall **16**. For example, as exemplified in FIG. **1**, transition member **22** extends between lower end **28** of sidewall **18** and outer end **30** of lower end wall **16**. If the cyclone separator is oriented as shown in FIG. **1** (it is in an upright orientation), then transition member **22** extends downwardly and inwardly.

(30) It will be appreciated that transition member **22** may have a variety of configurations. For example, transition member **22** may be a single surface that extends at an angle from lower end **28** of sidewall **18** to outer end **30** of lower end wall **16** (see for example FIGS. **15** and **16**). As exemplified, sidewall **18** extends between the first end **14** and transition member **22** in a first direction (vertically), second end **16** extends in a second direction (horizontally), and transition member **22** extends in at least one third direction from sidewall **18** to the second end **16**.

Accordingly, it will be appreciated that the transition member extends in a third direction (other than the direction of sidewall **18** and the direction of end wall **16**). Accordingly, transition member **22** may be at an angle to the longitudinal axis of sidewall **18**, at an angle to the longitudinal axis A of the cyclone separator itself and at an angle to the plane of lower end **16** (the horizontal plane as exemplified in FIGS. **15** and **16**).

(31) Alternately, a plurality of angled surfaces may be provided. Alternately, and preferably, transition member **22** may be curved and, more preferably, is radiused. In a particularly preferred embodiment, transition member **22** describes part of an arc of a circle which may have a radius from 0.125 inches to 2 inches, more preferably from 0.25 to 1 inch, even more preferably from about 0.375 to 0.75 inches and most preferably about 0.5 inches.

(32) Preferably, dirt outlet **24** is provided in a lower portion of the cyclone separator. Preferably, if transition member **22** is provided, at least a portion of dirt outlet **24** is provided in transition member **22**. For example, as exemplified in FIGS. **17** and **18**, dirt outlet **24** may extend above transition member **22**. Preferably, as exemplified in FIG. **1**, dirt outlet **24** is positioned completely within transition member **22**. For example, as exemplified in FIG. **14**, dirt outlet **24** extends to the juncture of transition member **22** and sidewall **18**. Alternately, as exemplified in FIG. **15**, dirt outlet **24** may terminate at a position below the juncture of transition member **22** and sidewall **18**.

(33) One or more dirt outlets **24** may be provided for a single cyclone chamber **26**. Preferably, a single dirt outlet **24** is provided, as exemplified in FIG. **1**. The one or more dirt outlets **24** may be positioned at any angular displacement B with respect to inlet **18** (see for example FIG. **21**). Dirt outlet may be displaced from about 90 to about 330 degrees, preferably from about 180 to about

300 degrees, more preferably from about 240 to about 300 degrees and most preferably about 270 degrees in the flow direction from air inlet **18**. For example, as shown in FIGS. **12** and **18**, dirt outlet **24** may be displaced about 180 degrees in the flow direction around cyclone chamber **26** from inlet **18** (i.e., dirt outlet **24** is generally opposed to air inlet **18**). Alternately, as exemplified in FIG. **17**, dirt outlet may be displaced about 90 degrees in the flow direction around cyclone chamber **26** from inlet **18**. It is preferred that outlet **24** having such an angular positioning is provided in a lower portion of the cyclone casing as discussed herein.

(34) As shown in FIG. **1**, the cyclone chamber has a height H (i.e., the distance between first and second opposed end walls **14** and **16**) and a diameter D (i.e. the diameter of sidewall **12**).

Accordingly, height H is the combined height of sidewall **18** and transition member **22**. In a broad aspect of this invention, height H and diameter D may be any of those known in the art. Preferably, height H is less than diameter D and, more preferably, height H is less than half of the diameter D.

(35) It will be appreciated that transition member **22** may have any desired length. Accordingly, transition member **22** may extend from end wall **16** to inlet **18**. However, it is preferred that a portion of sidewall **12** is provided between inlet **18** and transition member **24**. For example, as shown in FIG. **1**, sidewall **12** has a portion having a distance d that extends from bottom **13** of inlet **18** to lower end **28** of sidewall **18**. Preferably, this section of sidewall is straight and, more preferably parallel to axis A (e.g., vertical as exemplified in FIG. **1** wherein the cyclone separator is oriented with the longitudinal axis A of the cyclone extending vertically). In particular, it is preferred that the section of sidewall having a length d is parallel to the longitudinal axis A of cyclone chamber **26**. Lower end **13** of inlet **12** is preferably closer to second lower end **16** than first upper end **14**. Accordingly, distance d may be less than the vertical height of the inlet **12**.

(36) Dirt collection chamber or separated material collection chamber **32** is positioned in fluid flow communication with dirt outlet **24** and exterior to cyclone chamber **26**. Dirt outlet **24** is a lateral outlet, i.e., the outlet is provided in sidewall **18** or transition member **22** so that the separated material travels at least partially laterally as it exits cyclone chamber **26**. In the embodiment of FIG. **8** wherein the cyclone is inverted, the upper end (second end **16**) may be open. In such a case, the open end would function as the dirt outlet.

(37) Dirt collection chamber **32** may be of any shape. As shown in FIG. **2**, dirt or other heavy material will exit cyclone chamber **26** laterally via outlet **24** and then travel downwardly to accumulate on collection surface **34**. As exemplified, dirt collection chamber **32** has a collection surface **34**, sidewall **36** and a top wall **38**.

(38) Dirt collection chamber **32** surrounds at least a portion of the cyclone casing. Preferably, dirt collection chamber **32** surrounds the portions of the sidewall of the cyclone casing that are not provided with inlet **18**. Alternately, as exemplified in FIG. **21**, it may surround all of sidewall **12**. The sidewalls of the cyclone casing and the collection chamber are spaced apart so as to define a gap, that may be an annular gap G if it fully surrounds the cyclone casing, that has a width D. Accordingly, the cyclone casing may be positioned fully within dirt collection chamber **32** as exemplified in FIG. **1**.

(39) In addition, the surface of collection chamber **32** opposed to and facing second end **16** (collection surface **34** in the cyclone is oriented upright and opposed surface **33** if the cyclone is inverted) is spaced from end **16** of the cyclone casing. Preferably, collection surface **34** is an opposed surface facing the second end **16** and spaced from the second lower end **16** by at least about 0.5 inches.

(40) Preferably, sidewall **12** may have an inlet section and a second section, and fluid inlet **18** is provided on the inlet section and the portion of sidewall **36** of separated material collection chamber **32** facing the second section of sidewall **12** of the cyclone casing is spaced apart by at least about 0.5 inches. As exemplified a gap having a width $D_{sub.g1}$ may be provided around the second section of sidewall **12**. Distance $D_{sub.g1}$ is at least 0.5 inches. It will be appreciated that the annular gap $D_{sub.g1}$ may be larger and need not have a uniform width. Further, a gap having a

width D.sub.g2 may be provided around the inlet section of sidewall **12**. It will be appreciated that width D.sub.g2 may be less than 0.5 and may be zero such that wall **69** of passage **68** merges with sidewall **36** of collection chamber **32** (see for example the embodiment of FIG. **14**).

(41) It will be appreciated that if the cyclone casing is positioned completely within dirt collection chamber **32** (i.e. dirt collection chamber **32** extends completely around the cyclone separator as exemplified in FIG. **21**) then, top wall **38** of dirt collection chamber **32** may be annular in shape (i.e., extend between sidewall **12** of the cyclone and sidewall **36** of the dirt collection chamber **32**).

(42) As exemplified in FIG. **4**, top wall **38** of dirt collection chamber **32** and upper first end wall **14** of cyclone chamber **36** may be connected together and are preferably integrally molded as a continuous plate **40**. Preferably, air outlet **24** may be provided as part of plate **40** and may be integrally molded therewith.

(43) As exemplified in FIGS. **1** and **22**, cyclone separator assembly **10** may be provided as a second cyclonic stage **130**. Accordingly, cyclone separator assembly **10** preferably has provided as part thereof the air outlet or vortex finder **42** of an upstream cyclonic stage **128** (which is preferably a single cyclone but, as will be appreciated, could be a plurality of cyclones in parallel wherein the vortex finder of each is secured to, and removable with, cyclone separator assembly **10**).

Preferably, outlet **42** comprises a tubular member having sidewalls **44** wherein apertures **46** may be provided in a lower end thereof. Alternately, it will be appreciated that any air outlet or vortex finder of a cyclone known in the art may be utilized, and need not be provided as part of cyclone separator assembly **10**. Vortex finder **42** is preferably attached to the bottom of cyclone separator assembly **10**, e.g., bottom panel **50**, and may be molded as part of bottom panel **50**.

(44) If the cyclone is downstream from another filtration member or air treatment member, then air inlet **18** may be in airflow communication therewith by any means known in the art. In a particularly preferred embodiment as illustrated in FIGS. **1-5**, cyclone separator assembly **10** may incorporate an airflow passage from an upstream filtration stage to inlet **18**. As illustrated therein, outlet **42** is in airflow communication with header **48**. Header **48** is positioned between bottom panel **50** having an upper surface **52** and a lower surface **54** of collection surface **34**.

(45) If bottom panel **50** is openably, then in order to provide an airtight seal for bottom panel **50**, a sealing gasket **66**, O-ring or other sealing member known in the art may be provided. Gasket **66** may be mounted to, or removably mounted to, pivoting bottom **50**. In such a case, header **48** maybe positioned between gasket **66** and lower surface **54** of collection surface **34**. From header **48**, the air travels upwardly through side air passage **56** (see FIGS. **3**, **4** and **5**). It will be appreciated that air passage **56** may be of any configuration known in the art. The airflow path is exemplified in FIG. **2**.

(46) In order to permit dirt collection chamber **32** to be emptied, bottom panel **50** may be pivotally attached to cyclone separator assembly **10**. For example, flange **58** may be provided on bottom panel **50**. A mating flange may be affixed to sidewall **36** (see FIG. **7**). The flanges may be pivotally connected by any means known in the art. A latch may be provided on sidewall **36**, which engages flange **60**, which is provided on bottom panel **50**. Accordingly, when in the closed position (shown in FIG. **1**), the latch may engage flange **60**, thereby securing bottom panel **50** in position. When released, bottom panel **50** may pivot to an open position (as shown in FIG. **7**) permitting dirt collected in collection chamber **32** to be emptied. In such a case, a sealing gasket **66** or the like is preferably provided, e.g., mounted to upper surface **52** of panel **50**, so as to open with panel **50**. Any sealing member known in the art may be used.

(47) In order to permit dirt collection chamber **32** to be emptied, an opening is provided to access the interior of dirt collection chamber **32**. Accordingly, collection surface **34** may be moveably or removable mounted or a door provided to permit access to dirt collection chamber **32**. For example, collection surface **34** may be removable, as exemplified in FIG. **6**. Alternately, or in addition, collection surface **34** may be pivotally mounted. For example, as exemplified in FIGS. **19** and **20**, collection surface **34** is pivotally mounted to sidewall **36** by pivot pin **78**. See also for example, the

embodiment of FIG. 9.

(48) It will be appreciated that larger particulate matter may be collected in cyclone chamber **26**. Accordingly, in any embodiment disclosed herein, access may be provided to the interior of cyclone chamber **26** as well. For example, cyclone chamber **26** may be opened independently of collection chamber **32**. Alternately, cyclone chamber **26** may be opened when dirt collection chamber **32** is opened so that cyclone chamber **26** and collection chamber **32** may be emptied concurrently.

(49) Accordingly, it will be appreciated that it is preferred that cyclone chamber **26** and collection chamber **32** are each openable. For example, lower end **16** and collection surface **34** may be each moveably mounted and may be joined together so that cyclone chamber **26** and collection chamber **32** are concurrently opened. As exemplified in FIGS. **19** and **20**, bottom or second end wall **16** of cyclone chamber **26** may be secured or joined to collection surface **34**, such as by ribs **104** and constructed to be openable with collection surface **34**. The openable portion of each of cyclone chamber **26** and collection chamber **32** may be moveably mounted by any means known in the art. For example, they may be slideably or translatably mounted. Preferably, they are pivotally mounted. Alternately, they may be removably mounted, such as by means of a screw mount, a bayonet mount or securing members such as wing nuts.

(50) As exemplified, cyclone chamber is openable at the juncture of transition member **22** and sidewall **12**. Accordingly, when a latch or lock is released, collection surface **34** may be moved to the open position shown in FIG. **20** and accordingly, bottom **16** and transition member **22** are concurrently moved to the open position.

(51) In an alternate embodiment, cyclone chamber **26** may be inverted. In such a case, as exemplified in FIGS. **8-12**, collection surface **34** is spaced from opposed surface **33**. Collection surface **34** and the first portion of the cyclone casing are openable and, preferably concurrently openable. For example, they may be moveably mounted and connected together. They may be moveably mounted by any means known in the art. For example, they may be slideably or translatably mounted. Preferably, they are pivotally mounted. Alternately, they may be removably mounted, such as by means of a screw mount, a bayonet mount or securing members such as wing nuts. For example, as exemplified in FIGS. **11** and **12**, the bottom of the cyclone chamber (when the cyclone chamber is in an inverted orientation) and the collection chamber may both be emptied at the same time by the cyclone chamber and the dirt collection chamber having a bottom panel that are adjacent (e.g., lie in a common plane) and are preferably integrally formed.

(52) It will be appreciated that bottom **16** may be constructed to be opened subsequently to collection chamber **32** (e.g., bottom **16** may not be secured to collection surface **34**).

(53) If side air passage **56** is provided, then collection surface **34** may incorporate a cut out **62**, which meets, preferably in an airtight manner, with bottom **64** of side air passage **56**.

(54) An alternate embodiment is shown in FIGS. **8-12**. In this alternate embodiment, the cyclone is inverted. Accordingly, air inlet **18** and air outlet **20** are provided in first end **14**, which comprises the lower end of the cyclone separator apparatus **10**. Accordingly, in operation, air will enter inlet **18** via, e.g., tangential passage **68**. The air will then travel upwardly in a cyclonic fashion in cyclone chamber **26** prior to exiting through outlet **20**, which is preferably provided in first end **16**. Heavier material (e.g. particulate material) will exit outlet **24** provided, preferably, in transition member **22** and accumulate in the separated material collection chamber **32**. If a sealing gasket **66** is optionally provided as exemplified in FIG. **10**, then the upper surface of gasket **66** functions as collection surface **34**.

(55) In order to empty cyclone separator assembly **10**, a latch **72** may be provided. For example, button **70** of latch **72** may be pressed moving arm **74** outwardly thereby disengaging arm **74** from flange **60**, thereby permitting bottom panel **50** to pivot downwardly (as shown in FIGS. **11** and **12**) to permit the interior cyclone chamber **26** and collection area **22** to be emptied. Bottom panel **50** may be pivotally mounted to sidewall **36** of casing **76** such as by pivot pin **78** extending

transversely to connect flange **58** of bottom panel **50** with flange **80** affixed to sidewall **36**.

(56) Accordingly, by opening bottom panel **50**, preferably over a garbage can or the like, material collected inside cyclone chamber **26** and collection chamber **32** may be emptied. In this embodiment, the collection surface **34** and second end **16** are integrally formed (i.e. they are defined by bottom panel **50**). Accordingly, both chambers may be emptied concurrently. In an alternate design they may be separately opened (e.g., if collection surface **34** and second end **16** are separate members).

(57) In a particularly preferred embodiment, cyclone separator apparatus **10** comprises an upper portion of an upright vacuum cleaner. Accordingly, housing **76** may be provided with a handle **82** which may be connected to cyclone separator apparatus **10** by any means known in the art. Air inlet **18** may be in airflow communication with a dirty air inlet or the like of a vacuum cleaner, as a first or subsequent cleaning stage, via opening **84** in panel **50** and opening **86** in optional gasket **66**.

(58) Further alternate embodiments are exemplified in FIGS. **13-15**. In these alternate embodiments, a plurality of the cyclone separators **10** is provided. Preferably, the plurality of cyclone separators **10** is provided in parallel. More preferably, the plurality of cyclone separators **10** is provided as a second cleaning stage, more preferably a second cyclonic cleaning stage and most preferably a second cyclonic cleaning stage of parallel cyclones in a surface cleaning apparatus. Any number of cyclone separator apparatus **10** may be provided.

(59) Each cyclone chamber has its own dirt collection chamber **32** that is isolated from the collection chambers **32** of the other cyclone separators **10**. For example, as shown, dividing walls **106** separate collection chambers **32**. It will be appreciated that, in alternate embodiments, collection chambers **32** need not have common walls. Dividing walls **106** may be secured to extension **112** of sidewall **12** of **16** of cyclone chamber **26** (see for example, FIG. **14**) so as to function as a rib **104**, such that second end **16** and collection **34** open concurrently. Alternately, dividing walls **106** may be removable from engagement with extension **112** such that collection chamber **32** may be emptied independent of cyclone chamber **26**. Cyclone chamber **26** may be separately openable or may not be openable.

(60) As exemplified, outlets **20** of cyclone separators **10** are in airflow communication with a header **108** having, preferably, a single outlet **110**. Accordingly, a single flow passage may be provided downstream from the cyclonic cleaning stage exemplified in FIGS. **13-15**.

(61) As exemplified, optionally inlets **18** of cyclone separators **10** may be positioned adjacent each other (e.g., side by side). See also FIG. **14**. In such a case, collection chamber **32** may surround only the second section, and not the inlet section, of sidewall **12**.

(62) It will be appreciated that any of the alternate or optional configurations or features may be used single or in any particular combination or sub-combination with other configurations or features disclosed herein.

(63) It will be appreciated by those skilled in the art that various modifications and additions may be made in each or within the scope of the following claims. In particular, it will be appreciated that one or more cyclones as disclosed herein may be provided in an appliance, preferably in a surface cleaning appliance and, more preferably in a vacuum cleaner. The cyclones may be provided at any orientation and are preferably either inverted or in an upright orientation.

Claims

1. A surface cleaning apparatus comprising a cyclone separator assembly, the cyclone separator assembly comprising: (a) a cyclone casing defining a cyclone chamber having first and second spaced apart opposed ends, a fluid inlet, a fluid outlet, a sidewall and a separated material outlet, the first end including a first end wall of the cyclone chamber, the second end including a second end wall, the sidewall defining a cyclone chamber perimeter, and the cyclone chamber has a cyclone chamber length extending between the first and second end walls, wherein the fluid inlet

and the fluid outlet are provided at the first end and the separated material outlet comprises an opening formed in the sidewall at the second end; and, (b) a separated material collection chamber in communication with the separated material outlet and exterior to the cyclone chamber, the separated material collection chamber having a first end including a first end wall, a second spaced apart opposed end having a second end wall and a sidewall extending between the first and second ends of the separated material collection chamber, the separated material collection chamber has a separated material collection chamber length extending between the first and second end walls of the separated material collection chamber wherein the separated material collection chamber length is longer than the cyclone chamber length, wherein one of the end walls of the separated material collection chamber is spaced from and faces one of the end walls of the cyclone chamber and a support member extends therebetween, wherein the sidewall of the cyclone chamber along a first portion of the cyclone chamber perimeter extends continuously from the first end wall to the second end wall and is in contact with the second end wall, and the sidewall of the cyclone chamber along a remainder of the cyclone chamber perimeter has the separated material outlet, wherein a common end of the cyclone chamber and the separated material collection chamber are concurrently openable, and wherein at least a portion of the sidewall of the separated material collection chamber is spaced from and faces the sidewall of the cyclone chamber.

2. The surface cleaning apparatus of claim 1 wherein the common end is a dirt collection surface of the cyclone chamber and the separated material collection chamber.

3. The surface cleaning apparatus of claim 1 wherein the first end of the cyclone chamber is a lower end of the cyclone chamber and the second end of the cyclone chamber is an upper end of the cyclone chamber.

4. The surface cleaning apparatus of claim 3 wherein the sidewall of the separated material collection chamber surrounds a portion of the sidewall of the cyclone chamber that does not have the fluid inlet.

5. The surface cleaning apparatus of claim 4 wherein a first portion of the separated material collection chamber is positioned between the sidewall of the cyclone chamber and the sidewall of the separated material collection chamber and a second portion of the separated material collection chamber is positioned between one of the end walls of the separated material collection chamber and one of the end walls of the cyclone chamber.

6. The surface cleaning apparatus of claim 5 wherein the first portion of the separated material collection chamber is U-shaped.

7. The surface cleaning apparatus of claim 1 wherein in use to clean a floor, the fluid inlet and the fluid outlet are provided at a lower end of the cyclone chamber, the separated material outlet is provided at an upper end of the cyclone chamber and one of the end walls of the separated material collection chamber is spaced from and faces one of the end walls of the cyclone chamber, and wherein a first portion of the separated material collection chamber is positioned between the sidewall of the cyclone chamber and the sidewall of the separated material collection chamber and a second portion of the separated material collection chamber is positioned between the one of the end walls of the separated material collection chamber the one of the end walls of the cyclone chamber.

8. The surface cleaning apparatus of claim 1 wherein the sidewall of the separated material collection chamber surrounds a portion of the sidewall of the cyclone chamber that does not have the fluid inlet.

9. The surface cleaning apparatus of claim 7 wherein the first portion of the separated material collection chamber is U-shaped.

10. The surface cleaning apparatus of claim 1 wherein the first portion of the cyclone chamber perimeter has a first length and the remainder of the cyclone chamber perimeter has a remainder length that is shorter than the first length, and the separated material outlet extends across the remainder length.

11. A surface cleaning apparatus comprising a cyclone separator assembly, the cyclone separator assembly comprising: (a) a cyclone chamber having a cyclone axis of rotation, a fluid inlet and a fluid outlet provided at a first end wall of a first end of the cyclone chamber, a separated material outlet provided at a second opposed end of the cyclone chamber, which comprises a second opposed end wall of the cyclone chamber and a sidewall, the separated material outlet formed in the sidewall, wherein the second end wall has a cyclone chamber face which faces towards the first end wall and an opposed face and wherein the cyclone chamber has a cyclone chamber length in an axial direction and a cyclone chamber perimeter; and, (b) a separated material collection chamber in communication with the separated material outlet and exterior to the cyclone chamber, the separated material collection chamber having a sidewall extending between first and second opposed end walls of the separated material collection chamber and a separated material collection chamber length in the axial direction wherein the separated material collection chamber length is longer than the cyclone chamber length, wherein one of the end walls of the separated material collection chamber is spaced from and faces one of the end walls of the cyclone chamber and a support member extends therebetween, wherein the sidewall of the cyclone chamber extends along the cyclone chamber perimeter and at least a portion of the sidewall of the cyclone chamber extends from the first end wall to the second end wall and is in contact with the second end wall, wherein a common end of the cyclone chamber and the separated material collection chamber concurrently opens the cyclone chamber and the separated material collection chamber, and wherein at least a portion of the sidewall of the separated material collection chamber is spaced from and faces the sidewall of the cyclone chamber whereby a first portion of the separated material collection chamber is positioned between the sidewall of the cyclone chamber and the sidewall of the separated material collection chamber, and wherein a second portion of the separated material collection chamber is positioned between an end wall of the separated material collection chamber and the opposed face.

12. The surface cleaning apparatus of claim 11 wherein the common end is a dirt collection surface of the cyclone chamber and the separated material collection chamber.

13. The surface cleaning apparatus of claim 11 wherein the fluid inlet and the fluid outlet are provided at a lower end of the cyclone chamber and the separated material outlet is provided at an upper end of the cyclone chamber.

14. The surface cleaning apparatus of claim 12 wherein the sidewall of the separated material collection chamber surrounds a portion of the sidewall of the cyclone chamber that does not have the fluid inlet.

15. The surface cleaning apparatus of claim 11 wherein the first portion of the separated material collection chamber is U-shaped.

16. The surface cleaning apparatus of claim 11 wherein the sidewall of the separated material collection chamber surrounds a portion of the sidewall of the cyclone chamber that does not have the fluid inlet.

17. The surface cleaning apparatus of claim 16 wherein the first portion of the separated material collection chamber is U-shaped.

18. The surface cleaning apparatus of claim 11 wherein the portion of the sidewall of the cyclone chamber has a first length along the cyclone chamber perimeter and a remaining portion of the sidewall of the cyclone chamber has a remaining length along the cyclone chamber perimeter that is shorter than the first length, and the separated material outlet is formed in the remaining portion of the sidewall of the cyclone chamber and extends across the remaining length.
